

OLIST MARKETPLACE ANALYTICS

Understanding Performance, Customer Satisfaction & Operational Priorities

Turning Delivery Reliability Into Customer Experience

Data Analytics Hackathon Submission | Olist Brazilian E-Commerce Dataset (Sep 2016 – Oct 2018)

1. Executive Summary

Delivery reliability is the strongest observable driver of customer dissatisfaction on the Olist marketplace. Orders delivered after their estimated delivery date received an average review score of 2.26★, compared with 4.28★ for orders delivered early — a gap confirmed by a Mann-Whitney test ($p < 0.0001$, rank-biserial correlation ≈ -0.64 , a large effect). Late orders generated 62.8% 1–2★ reviews versus 9.5% for early orders.

The relationship intensifies with delay severity: the 1–2★ rate rises from 32.5% for orders 1–3 days late to 68.0% at 4–7 days late and roughly 78–80% beyond a week late. A logistic regression that controls for shipping distance, freight cost, order value, cross-state shipping, and customer state confirms the effect is not confounded: a late order carries roughly 16 times the odds of a low review versus an early order ($OR = 16.4$, $p < 0.001$), and the result holds under a stricter definition of dissatisfaction (1★ only, $OR = 16.4$, $p < 0.001$).

This is not a story of a handful of underperforming sellers. Delivery reliability deteriorated across the marketplace in a specific window: the late-delivery rate rose from 3.83% (Aug–Oct 2017) to 16.63% (Feb–Mar 2018), while average review scores fell from 4.246★ to 3.830★ (Mann-Whitney $p \approx 3 \times 10^{-105}$). March 2018 alone recorded an 18.96% late-delivery rate — the highest of any month in the dataset. Seller-level checks show this spike was broad-based: the ten sellers with the most late orders in March 2018 accounted for only about 20% of that month's late volume.

Freight cost is a secondary, independent contributor: higher freight value is associated with higher odds of a low review even after controlling for lateness ($OR \approx 1.69$, $p < 0.001$), though the effect is far smaller than delivery status. Distance, order value, cross-state shipping, and customer state show no significant independent effect once delivery status and freight are accounted for — dissatisfaction is concentrated in specific delivery outcomes and specific segments, not spread evenly across geography.

Intervention should be targeted rather than uniform. The clearest priority segment is Rio de Janeiro (RJ) customers ordering Bed/Bath/Table products: 1,333 orders, a 15.1% late-delivery rate, and 147 late orders rated 1–2★. High-volume categories — Bed/Bath/Table, Health & Beauty, Sports & Leisure, and Computers & Accessories — carry the largest absolute burden of delivery-linked dissatisfaction and should anchor operational focus.

Priority actions for Olist leadership

- 1) Prevent — build proactive monitoring of orders approaching their estimated delivery date without confirmed delivery.
- 2) Prioritize — target intervention at high-burden state×category segments (led by RJ × Bed/Bath/Table).
- 3) Escalate — investigate freight-cost burden and individual seller performance as secondary levers, recognizing that the 2018 reliability crisis was systemic rather than seller-specific.

These findings are drawn from observational data. They establish strong, statistically robust associations and a clear set of intervention priorities, but they do not prove that late delivery is the sole cause of poor reviews; the interpretation throughout this report is deliberately correlational.

2. Business Problem & Objectives

Olist connects small and medium Brazilian merchants to major online marketplaces through a single contract. Sellers fulfill orders through Olist's logistics partners, and customers receive a post-delivery satisfaction survey (1–5 stars, with optional comments). Leadership holds roughly two years of order history but has lacked a single, consolidated view connecting delivery performance, seller and geographic patterns, category economics, payment behavior, and review outcomes.

This analysis independently investigates what is actually shaping customer experience on the platform and where the clearest opportunities exist to improve it as Olist continues to scale. It addresses the six core questions set out in the hackathon problem statement: marketplace performance over time, delivery performance and satisfaction, seller and geographic

patterns, product category performance, payment behavior, and root-cause analysis of low review scores. Findings are evidence-based; no causal claim is made that observational data cannot support.

3. Analytical Approach

The analysis joins nine source tables — orders, order items, payments, reviews, customers, products, sellers, geolocation, and category translations — into an order-level analytical base, since orders are the natural unit for a review outcome.

Delivery and review measurement

Delivery delay was computed only for the 96,476 delivered orders (of 99,441 total) as the difference, in whole calendar days, between the actual and estimated delivery dates. Orders were classified as Early, On-time, or Late. Non-delivered orders (shipped, canceled, unavailable, invoiced, processing, created, approved) were excluded from delay-based analysis because they lack a completion timestamp by construction — including them would bias delay measurement toward incomplete orders.

Review scores were aggregated to one value per order (mean of any duplicate review records; 555 orders had more than one) and merged one-to-one onto the delivered-order population.

Category, geography, and seller attribution

Products were translated to English category names and mapped to orders; orders with missing or multiple categories across their line items (2,074 orders) were excluded from category-level analysis to keep category attribution unambiguous, leaving 94,402 orders. Geolocation was aggregated to mean latitude/longitude per zip-code prefix (19,015 unique prefixes) before joining to avoid multiplying rows, and Haversine straight-line distance between customer and seller was calculated from these coordinates (available for 93,932 of 94,402 orders). Seller-level performance analysis was restricted to the 97,388 orders fulfilled by a single seller, excluding 1,278 multi-seller orders where responsibility cannot be cleanly attributed.

Statistical approach

Review scores are ordinal and non-normally distributed, so the Mann-Whitney U test (with rank-biserial correlation as effect size) was used to compare early- versus late-delivery review distributions, and to compare a 2017 baseline period against the 2018 deterioration period. Logistic regression was used to test whether delivery status remains associated with the odds of a low review (≤ 2 ★) after controlling for log-transformed distance, freight value, and order value, cross-state shipping, and customer state fixed effects. A robustness check re-ran the same model against a stricter target (1 ★ only).

4. Data Quality & Preparation

The dataset carries genuine real-world data-quality characteristics that were handled deliberately rather than ignored:

- Non-delivered orders (2,965 of 99,441) lack complete delivery timestamps and were excluded from delay-based metrics rather than imputed.
- Orders with more than one review record (555) were aggregated by mean score rather than arbitrarily deduplicated.
- Orders with more than one item (13,984) or payment record (4,446) were aggregated to the order grain (summed item/freight value; summed payment value) before analysis, rather than counted as raw rows.
- The geolocation table (1,000,163 rows, ~52 readings per zip prefix on average) was aggregated to one mean coordinate per prefix before any join, avoiding row multiplication.
- Category attribution required a single, unambiguous category per order; 2,074 orders with missing or mixed categories were excluded from category-level findings only (not from delivery or marketplace-level findings).
- Seller-level performance used single-seller orders only (97,388 of 99,441); multi-seller orders were excluded from that specific analysis to avoid misattributing performance.
- A small number of orders (about 470) could not be matched to both a customer and seller coordinate and were excluded from distance-based analysis only.

One internal inconsistency is worth flagging transparently: the count of March 2018 late orders differs slightly across sub-analyses (1,328 in the full delivered-order population versus approximately 1,303–1,306 in the seller-concentration check). This is a product of different, deliberately-scoped populations — the seller check necessarily excludes multi-seller orders — not a data error, but the report treats the full-population figure (1,328; 18.96% late rate) as authoritative for marketplace-wide claims.

5. Marketplace Performance Over Time

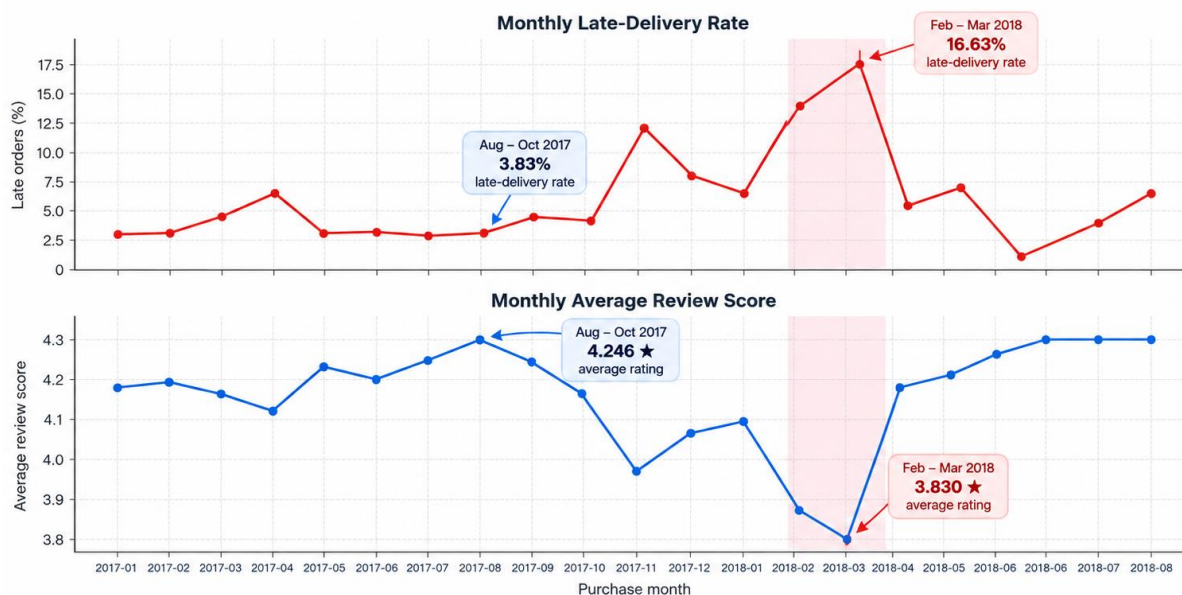
Order volume grew substantially over the observed period, from a handful of orders in late 2016 to a stable run rate of 6,000–7,500 delivered orders per month through 2018, with a pronounced peak around November 2017 (7,544 orders) that coincides with Black Friday seasonality. Average order value fluctuated in a comparatively narrow band (roughly R\$147–R\$185) and did not trend sharply up or down, suggesting growth was driven primarily by volume rather than rising basket size.

Growth and satisfaction did not move together throughout the period. For most of 2017, late-delivery rates stayed low (roughly 3–7%) and average review scores stayed high (around 4.2 ★). That relationship broke down in the winter of 2017–18: the late-delivery rate climbed to 12.4% in November 2017, eased slightly, then surged to 14.1% in February 2018 and 19.0% in March 2018 — the two highest months in the dataset — while average review scores fell correspondingly to 3.83–3.87 ★, the lowest sustained readings in the series. Reliability and scores both recovered by April–June 2018 (late rates back near 1–6%, scores back above 4.1 ★).

The comparison between a pre-crisis window (Aug–Oct 2017) and the crisis window (Feb–Mar 2018) is stark and statistically confirmed: the late-delivery rate rose from 3.83% to 16.63% (+12.8 percentage points, a 334% relative increase), the count of late orders rose from 491 to 2,255 (+359%), and the average review score fell from 4.246 ★ to 3.830 ★ (Mann-Whitney $p \approx 3 \times 10^{-105}$). Order volume and revenue kept growing through this window — growth did not pause while reliability deteriorated, which is precisely the scenario leadership needs visibility into as the platform continues to scale.

Delivery Reliability Deteriorated Sharply in Late 2017 – Early 2018

Late-delivery rates surged while average customer ratings declined.



Late-delivery rate
3.83% → 16.63%
(+12.8 percentage points, +334%)

Late orders (count)
491 → 2,255
(+359%)

Average review score
4.246 → 3.830
(–0.416 ★)

**A clear deterioration
in customer experience
coincided with a period
of marketplace expansion.**

Figure 1. Monthly late-delivery rate and average review score, 2017–2018. Reliability and satisfaction move together and both deteriorate sharply in the Feb–Mar 2018 window.

6. Delivery Performance & Customer Satisfaction

Delivery timing is measured as the gap, in days, between actual and estimated delivery date. The relationship to review scores is large, consistent, and statistically robust:

Delivery outcome	Orders	Avg. review score	1–2★ rate	1★ rate
Early	88,649	4.28	9.5%	6.78%
Late	6,535	2.26	62.8%	54.08%

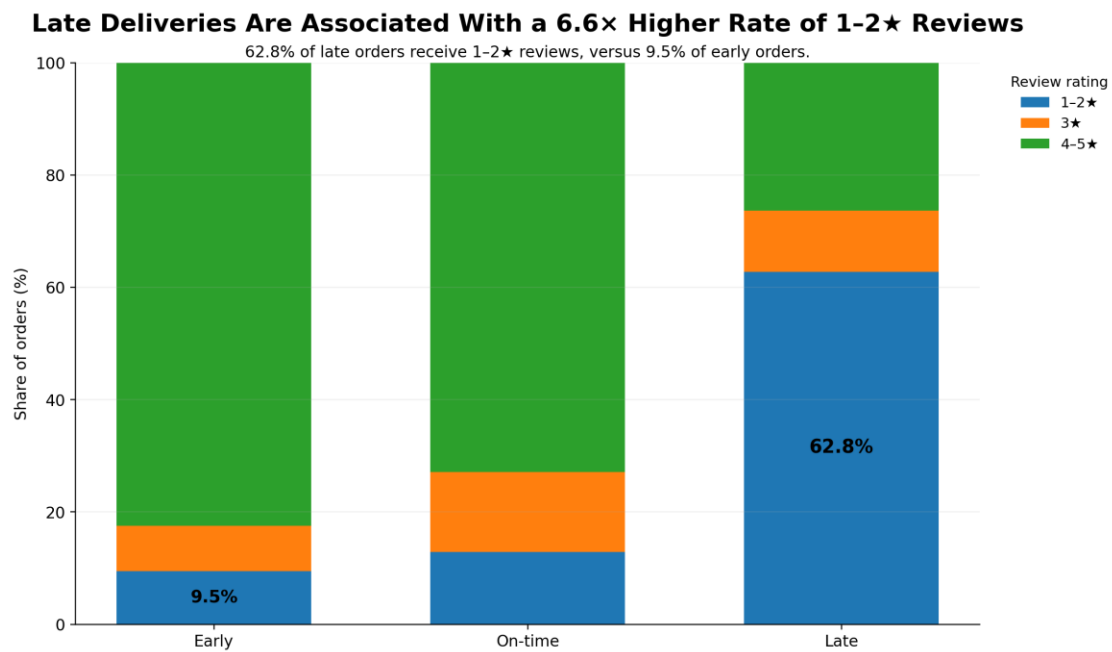


Figure 2. Review-rating composition by delivery status. Late orders show a 6.6× higher rate of 1–2★ reviews than early orders.

The Mann-Whitney U test comparing early and late review-score distributions is significant well beyond conventional thresholds ($U = 475,003,963.5$, $p < 0.0001$), and the rank-biserial correlation of -0.64 indicates a large effect — not a marginal statistical artifact of a large sample.

The relationship is dose-dependent: dissatisfaction rises with the length of the delay rather than switching on uniformly once an order is late.

Delay severity	1–2★ review rate
1–3 days late	32.5%
4–7 days late	68.0%
8–14 days late	80.3%
15+ days late	78.4%

Customer Dissatisfaction Accelerates as Delivery Delays Grow

1–2★ review rate rises sharply once orders become several days late.

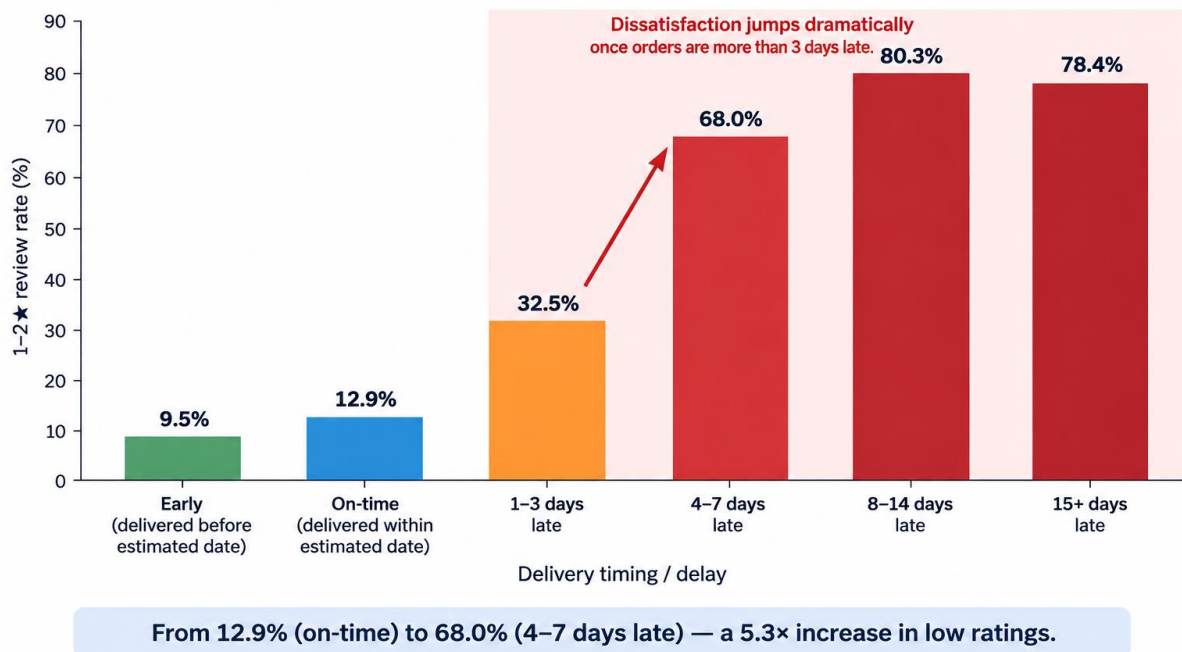


Figure 3. 1–2★ review rate by delay severity. Dissatisfaction rises sharply once an order is more than three days late, then plateaus at a high level.

The pattern holds across shipping distance: at every distance band, early deliveries score far higher than late ones (for example, under 100 km: 4.35★ early vs. 2.76★ late; 100–250 km: 4.38★ early vs. 2.49★ late). This rules out distance itself as the explanation for the delivery-satisfaction relationship — it is delivery outcome, not shipping distance, that drives the gap. The relationship also holds separately in 2017 and 2018 (1–2★ rate for late orders: 61.9% in 2017, 63.3% in 2018), confirming it is a stable structural feature of the marketplace rather than a one-time artifact of the 2018 crisis.

Late orders punch above their weight in the overall dissatisfaction pool: although late orders are a minority of delivered volume (6.8%), they account for 36.6% of all 1★ reviews and 32.4% of all 1–2★ reviews platform-wide.

Language note: these results describe a strong association — not proof that lateness alone causes a poor review. The root-cause section below tests whether the relationship survives statistical controls.

7. Root Causes of Customer Dissatisfaction

A logistic regression was fit on 93,932 orders with a low-review flag (≤ 2 ★) as the outcome, controlling simultaneously for delivery status, log-transformed shipping distance, log-transformed freight value, log-transformed order value, whether the shipment crossed a state line, and customer-state fixed effects.

Primary driver

Delivery status dominates the model. Relative to an early delivery, a late delivery carries roughly 16 times the odds of a low review (OR = 16.4, $p < 0.001$); even an on-time delivery (arriving on, not before, the estimated date) carries modestly higher odds than an early one (OR = 1.44, $p < 0.001$). This finding is robust to a stricter definition of dissatisfaction: rerunning the model against 1★-only reviews produces an almost identical late-order odds ratio (OR = 16.4, $p < 0.001$) and on-time odds ratio (OR = 1.38, $p = 0.0017$).

Secondary contributor

Freight value is independently associated with dissatisfaction after controlling for delivery status and distance: each unit increase in log freight value raises the odds of a low review by roughly 69% ($OR \approx 1.69$, $p < 0.001$), and the same pattern appears in the 1★-only robustness check ($OR \approx 1.70$, $p < 0.001$). A simple freight-share view corroborates this at a smaller scale: the low-review rate rises from 12.7% for orders where freight is under 10% of order value to 15.6% where freight exceeds 50% of order value, alongside a rising late-delivery rate (6.9% to 8.4%) across the same buckets. Freight burden is a real but secondary factor — an order of magnitude smaller in effect than delivery status.

Contextual / non-significant factors

Shipping distance, order value, whether a shipment crossed a state line, and customer state each fail to reach statistical significance once delivery status and freight are controlled for (all $p > 0.05$ in the primary model). This is an important and non-obvious finding: the raw gap between cross-state orders (14.2% low-review rate) and same-state orders (10.5%) that appears in a simple cross-tabulation does not survive controlling for delivery outcome and freight cost — it appears to be a downstream consequence of cross-state shipments being more prone to delay and higher freight cost, rather than an independent driver in its own right.

Why this isn't a contradiction with the geographic findings

The regression tests a general customer-state effect and finds none of the 26 states/DF individually significant. The state×category segment analysis in Section 8, by contrast, looks at specific combinations (e.g., RJ paired with Bed/Bath/Table) and at raw, unadjusted volumes of dissatisfaction rather than a controlled odds ratio. The two results are consistent: geography alone doesn't move the needle, but specific state×category combinations concentrate enough volume and delay risk to be worth targeting operationally.

No single factor should be treated as the sole cause of low reviews. Delivery lateness is overwhelmingly the primary driver by evidence strength and magnitude; freight cost is a real secondary contributor; distance, order value, and broad geography do not add explanatory power once those two are accounted for.

8. Seller & Geographic Patterns

Individual sellers show meaningful heterogeneity in outcomes: among sellers with at least 100 orders, late rates and average review scores vary considerably (for example, one 227-order seller shows a 42-order late count and a 3.51★ average, while a 388-order seller shows 74 late orders and a 3.99★ average). This heterogeneity is real and supports ongoing seller-level performance monitoring.

However, the 2018 reliability crisis itself was not concentrated among a small number of underperforming sellers. In March 2018 — the worst month in the dataset — the ten sellers with the most late orders accounted for only about 20% of that month's total late-order volume among single-seller-attributable orders. This indicates a systemic, marketplace-wide breakdown in fulfillment or logistics during that window rather than a seller-quality problem that could be solved by managing a handful of accounts.

March 2018 Late Deliveries Were Broadly Distributed Across Sellers

Top 10 sellers contributed only 20.1% of 1,306 March late orders.

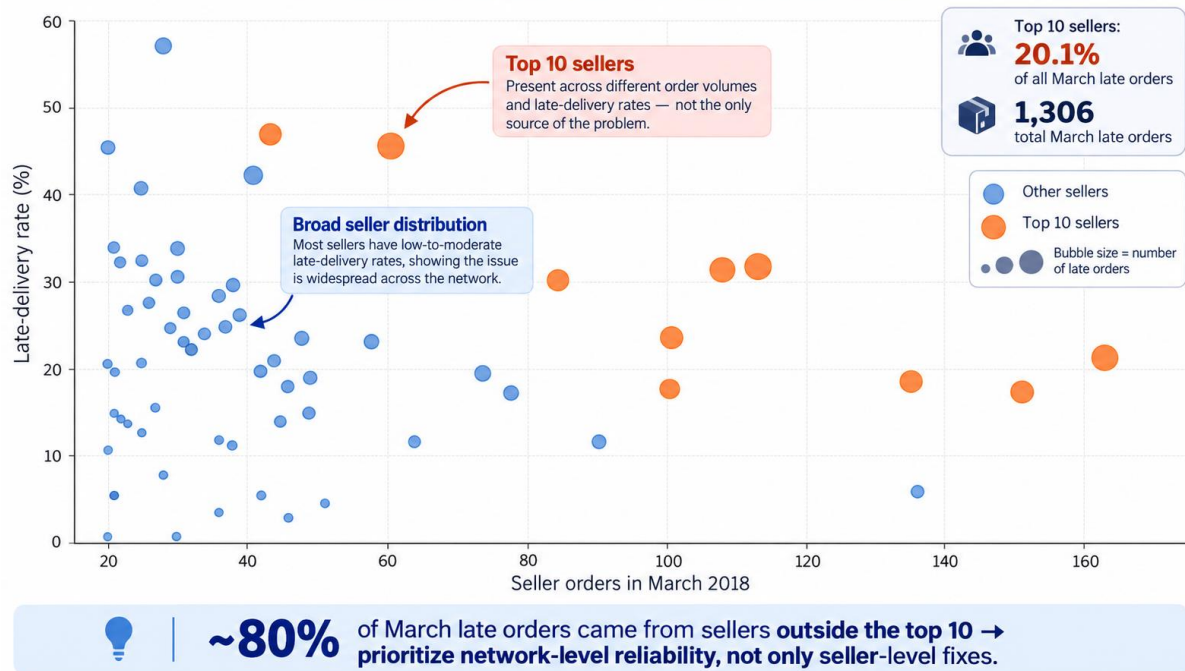


Figure 4. March 2018 seller-level late-delivery rates. Elevated late rates are spread broadly across the seller base rather than concentrated in a few accounts — the top 10 sellers contributed only ~20% of that month's late orders.

Geographically, the clearest actionable pattern is not a broad state effect (as shown in Section 7) but specific state×category combinations that combine meaningful order volume with elevated late-delivery risk and a substantial count of resulting low ratings:

Segment	Orders	Late rate	Late orders rated 1–2★
RJ × Bed/Bath/Table	1,333	15.1%	147
SP × Bed/Bath/Table	4,260	~5%	~102
SP × Health & Beauty	3,678	~4–5%	~103
RJ × Sports & Leisure	882	~14%	~90
RJ × Computers & Accessories	824	~12.5%	~81

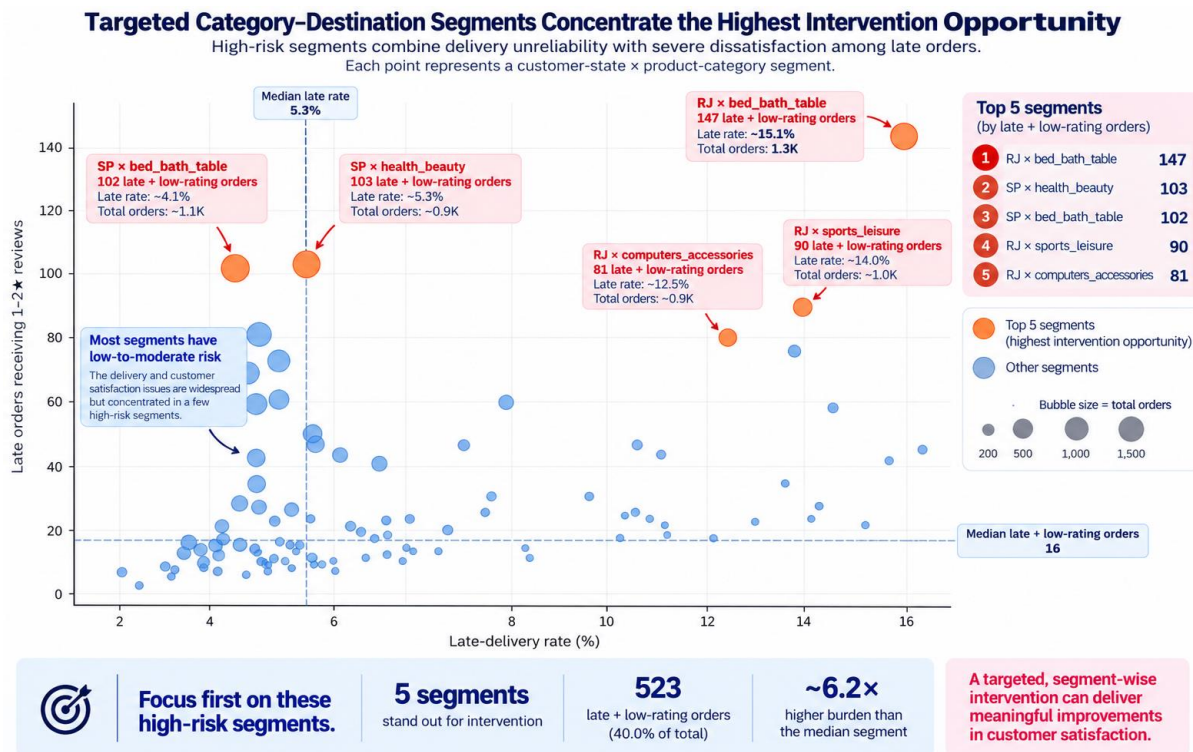


Figure 5. State×category segments ranked by late-delivery rate and resulting low-rating volume. The five highlighted segments concentrate a disproportionate share of delivery-linked dissatisfaction.

RJ × Bed/Bath/Table stands out as the single highest-priority segment: it combines an above-average late rate (15.1%, versus roughly 6.8% marketplace-wide) with the largest absolute count of resulting 1–2★ reviews of any segment examined. The São Paulo segments carry lower late rates but very high volume, so their absolute low-rating burden is comparably large — a reminder that high-volume, average-reliability segments can matter as much as high-risk, lower-volume ones.

9. Product Category Performance

Among categories with at least 500 orders, a small group of high-volume categories combine substantial order counts with a correspondingly large absolute number of orders rated 1★ following a late delivery. Ranked by that combined burden: Bed/Bath/Table (9,075 orders), Health & Beauty (8,579), Sports & Leisure (7,463), Computers & Accessories (6,479), Furniture & Décor (6,105), Watches & Gifts (5,453), and Housewares (5,641).

These are the categories most strategically important to prioritize — not because their per-order dissatisfaction rate is exceptional, but because their scale means that even a marketplace-average delivery failure rate translates into a large absolute number of dissatisfied customers. This distinguishes them from smaller categories that might show a worse average rate on a much smaller base and would be a lower-value target for intervention.

10. Payment Behavior

Credit card is the dominant payment method among delivered orders (72,824), followed by boleto (19,191), voucher (2,977), and debit card (1,483). Credit card purchases carry meaningfully more installments on average (3.55) than boleto or debit card, which structurally only support single-installment payment.

Average review scores differ only marginally across payment types (in the 4.10–4.23★ range per the underlying analysis) — a gap an order of magnitude smaller than the 2-point spread between early and late deliveries. Payment method and installment choice do not appear to be a meaningful lever for customer satisfaction on this platform. A genuine null finding is

still a useful finding here: it tells Olist that payment-experience initiatives are unlikely to move satisfaction and that operational focus is better spent on delivery reliability.

11. Key Insights

Insight 1 — Delivery lateness is the dominant, statistically robust driver of dissatisfaction

What we found: Late orders average 2.26★ versus 4.28★ for early orders, with a 62.8% vs. 9.5% gap in 1–2★ rates.

Evidence: Mann-Whitney U test, $p < 0.0001$, rank-biserial ≈ -0.64 ; confirmed in a controlled logistic regression, $OR = 16.4$, $p < 0.001$, robust to a 1★-only definition.

Why it matters: Delivery reliability is not one factor among many — it is the single largest lever available to move satisfaction outcomes.

Where it matters most: Effect strengthens with delay length (32.5% $\rightarrow$ ~80% low-review rate) and holds across all distance bands and both years observed.

Implication: Any customer-experience strategy that does not center on delivery reliability is very unlikely to move the needle materially.

Insight 2 — The marketplace suffered a systemic, dateable reliability breakdown, not a seller-quality problem

What we found: Late-delivery rate rose from 3.83% (Aug–Oct 2017) to 16.63% (Feb–Mar 2018); March 2018 alone hit 18.96%, the worst month in the dataset.

Evidence: Mann-Whitney test on review scores across the two windows, $p \approx 3 \times 10^{-105}$; the top 10 sellers by late-order count in March 2018 explain only ~20% of that month's late volume.

Why it matters: A systemic breakdown requires a systemic, operational fix (capacity, carrier performance, estimated-delivery-date calibration) — not seller-by-seller account management.

Where it matters most: Concentrated in the Nov 2017 – Mar 2018 window; the marketplace recovered by mid-2018.

Implication: Olist should build the operational capability to detect this kind of marketplace-wide deterioration in near-real time, rather than discovering it after the fact in a retrospective analysis.

Insight 3 — Freight cost is a real, if secondary, independent contributor

What we found: Freight value predicts low reviews even after controlling for delivery status, distance, and order value.

Evidence: $OR \approx 1.69$ per unit increase in log freight value, $p < 0.001$; consistent in the 1★-only robustness check ($OR \approx 1.70$).

Why it matters: Freight burden and delivery reliability may compound — customers who pay a high freight premium may hold a higher bar for on-time delivery.

Where it matters most: Orders where freight exceeds 40–50% of order value (low-review rate 15.6% vs. 12.7% for the lowest freight-share band).

Implication: Worth a dedicated logistics-cost review, but should not be prioritized ahead of delivery-reliability fixes given the much smaller effect size.

Insight 4 — Dissatisfaction is concentrated in identifiable, high-volume segments

What we found: RJ × Bed/Bath/Table (1,333 orders, 15.1% late rate, 147 low-rated late orders) is the clearest single intervention priority; Bed/Bath/Table, Health & Beauty, Sports & Leisure, and Computers & Accessories are the categories carrying the largest absolute dissatisfaction burden marketplace-wide.

Evidence: State×category segment ranking (Section 8) and category-level 1★ burden ranking (Section 9).

Why it matters: A targeted intervention on a small number of segments can address a disproportionate share of the dissatisfaction volume without a costly uniform overhaul.

Where it matters most: Rio de Janeiro deliveries for physically bulky, home-goods categories in particular.

Implication: Operational and logistics review should start with these named segments, not a generic nationwide rollout.

12. Strategic Recommendations

Priority	Action	Target	Evidence	Expected outcome	KPI
1	Build proactive delivery-risk monitoring that flags orders approaching their estimated delivery date without a confirmed delivery, enabling early carrier/seller follow-up.	All in-transit orders, marketplace-wide	OR = 16.4 for late vs. early delivery on low-review odds; effect strengthens with delay length (32.5%–80% low-review rate)	Reduction in late-delivery rate and in the share of long (4+ day) delays	Late-delivery rate (%); 1–2★ review rate
2	Target operational and logistics review at the highest-burden state×category segments, led by RJ × Bed/Bath/Table, and at the highest-volume categories overall (Bed/Bath/Table, Health & Beauty, Sports & Leisure, Computers & Accessories).	RJ × Bed/Bath/Table and named high-volume categories	Segment ranking by late orders resulting in 1–2★ reviews (Section 8); category-level 1★ burden ranking (Section 9)	Reduced concentration of dissatisfaction in the highest-burden segments	Low-review rate within targeted segments vs. marketplace average
3	Review freight pricing/logistics cost structure for high-freight-share orders, and maintain ongoing seller-level performance monitoring independent of the systemic 2018 issue.	Orders with freight >40% of order value; individual sellers with persistently elevated late rates	Freight OR ≈ 1.69 after controlling for delivery status; seller-level late-rate heterogeneity (Section 8)	Modest further reduction in low-review rate beyond delivery-reliability gains	Low-review rate by freight-share bucket; seller-level late rate distribution

This order reflects impact, evidence strength, and feasibility: Priority 1 addresses the factor with by far the largest, most statistically robust effect and the clearest operational lever (an early-warning system); Priority 2 concentrates limited operational resources where the dissatisfaction volume is largest; Priority 3 addresses smaller, secondary effects that are still worth pursuing but should not compete for resources ahead of the first two.

A Targeted Reliability Strategy Can Turn the Analysis Into Action

From insight to impact: detect risks early, focus on what matters, and take targeted action.

RELIABLE
MARKETPLACES
HAPPIER
PEOPLE

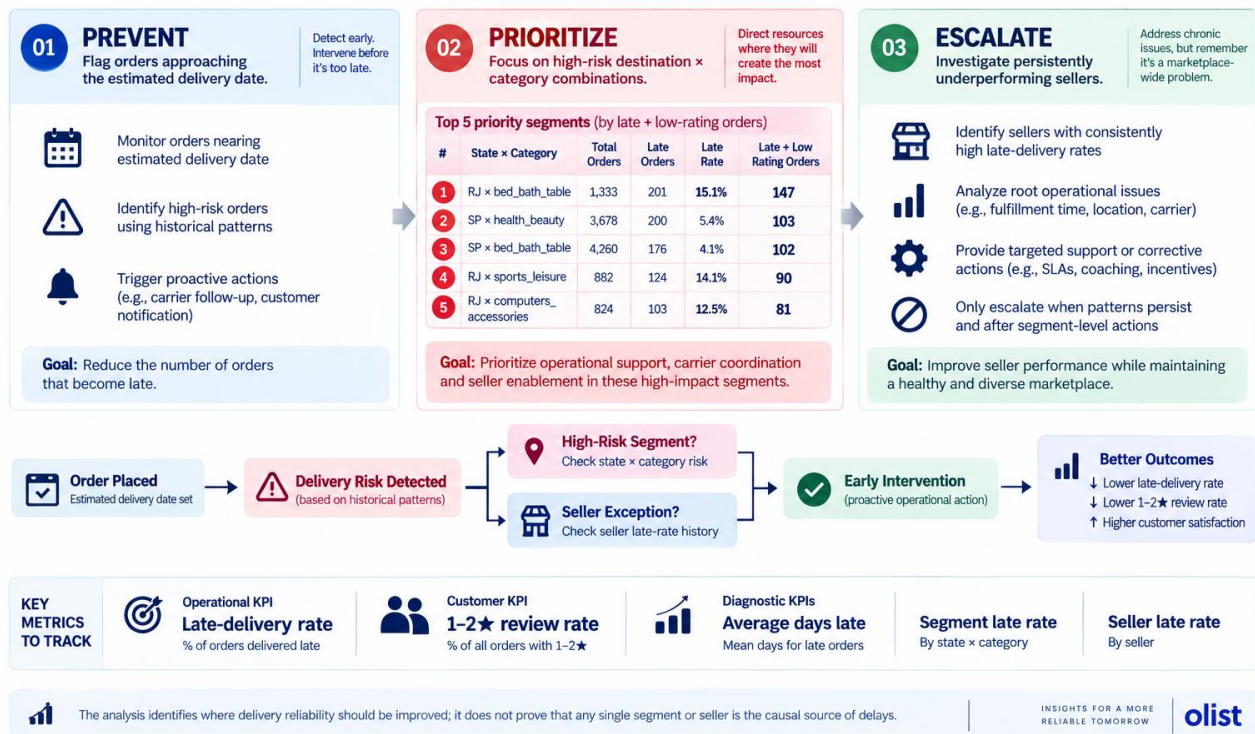


Figure 6. Recommended operating model: Prevent → Prioritize → Escalate, mapped to the top five priority segments and the KPIs leadership should track.

A Targeted Reliability Strategy Can Turn the Analysis Into Action

Use early detection, targeted intervention, and seller exception management.

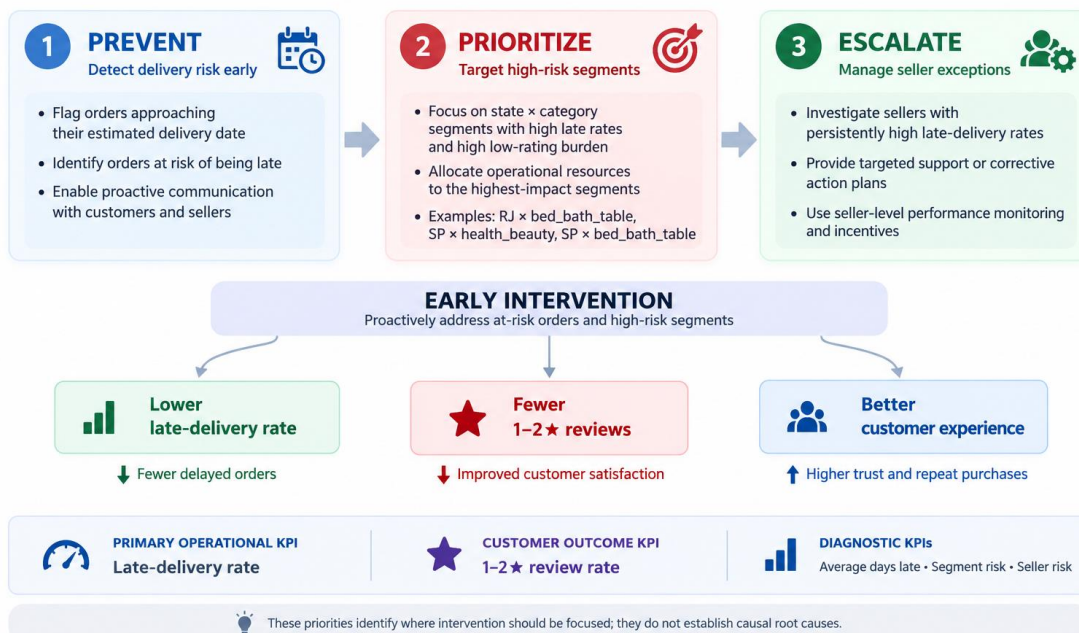


Figure 7. Summary view of the same three-step operating model and its expected outcomes.

13. Limitations

- This is observational data. Strong, statistically robust associations were established between delivery outcomes and review scores, but the data does not support a definitive causal claim; unobserved factors (e.g., product quality, listing accuracy, customer expectations) may also contribute.
- The regression's customer-state fixed effects were individually non-significant, likely reflecting limited per-state sample sizes for smaller states rather than proof that geography never matters — the segment-level findings in Section 8 show geography does matter in combination with category.
- Shipping distance was estimated as a straight-line (Haversine) distance between aggregated zip-code-prefix coordinates, not actual road or logistics-route distance, and was unavailable for about 470 orders.
- Category-level findings exclude 2,074 orders with missing or multiple product categories; seller-level findings exclude 1,278 multi-seller orders. Both exclusions are necessary for clean attribution but slightly reduce the analyzed population for those specific sections.
- A minor discrepancy in March 2018 late-order counts across sub-analyses (1,328 vs. ~1,303–1,306) reflects different population scopes, not a data error; see Section 4.
- Review-comment text and repeat-customer (`customer_unique_id`) behavior were not analyzed in depth in the underlying notebook and are not covered by this report; they remain open, optional directions for further work.
- The dataset spans September 2016 to October 2018 only. Current Olist operations, carriers, and seller base may differ from what is reflected here.

14. Conclusion

The marketplace grew steadily in order volume through the observed period, but that growth was accompanied by a real, dateable breakdown in delivery reliability in the winter of 2017–18 that measurably depressed customer satisfaction. Delivery lateness is, by a wide margin, the strongest and most robust factor associated with low review scores on this platform — a finding that survives statistical controls for distance, freight cost, order value, and geography, and that replicates under a stricter definition of dissatisfaction.

Freight cost contributes independently but at a much smaller scale. Payment behavior shows essentially no relationship to satisfaction. Geography alone does not predict dissatisfaction, but specific state×category combinations — above all, Rio de Janeiro customers ordering Bed/Bath/Table products — concentrate enough volume and risk to justify targeted intervention, alongside the platform's highest-volume categories.

The single highest-leverage action available to Olist is building the operational capability to detect and intervene on delivery risk before it reaches the customer, informed by where the burden concentrates. Everything else examined in this analysis — freight cost, seller-level variation, payment behavior, broad geography — is secondary to that central fact.